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MY SUMMARIES

IN MICROBIOLOGY

I have approximately twelve years of experience in the fields of Microbiology and Mycology, working in major areas such as Health Sciences (where I started my career) and Life Sciences (the area where I have greater expertise and focus). My work in the field of Mycology is the result of a trajectory built over eight years of continuous work in the area, in partnership with leading professionals working in Brazil, such as Aristóteles Góes-Neto [1], Elisandro Ricardo Drechsler-Santos [2], João Paulo Machado de Araújo [3], José Carmine Dianese [4], Maria Alice Neves [5] and Robert W. Barreto [6] (all mycologists of national and international renown). Over this time, I have also established research partnerships with international scientists, such as Charles W. Barnes [7], Mary Catherine Aime [8], Harry C. Evans [9] and Dominik Begerow [10]. I consider all of them my direct and indirect mentors in the field of Mycology. My areas of interest are genetics, taxonomy, ecology and evolution of microorganisms, focusing mainly on issues related to wild fungi and bacteria (organisms without commercial interest), pathogens of agricultural and environmental interest.

I have deep knowledge in analysis using biostatistical and bioinformatic techniques (see section Bioinformatics), with a focus on analyses of genomic and metagenomic data.

I am passionate and driven by scientific challenges and the discovery of the mysteries of the natural world. Visit the Publications section to see details of the scientific trajectory I have followed.

AND IN BIOINFORMATICS

I am a self-taught bioinformatician, with over ten years of experience in analyzing biological data using biostatistics and bioinformatics techniques. In recent years I have focused my attention and energy on building high-performance and/or high-availability bioinformatics tools, adapted to the highly connected web environment.

Today, I work professionally in modeling, architecture, development and coordination of multidisciplinary technology teams to create high-performance and high-availability biological data processing environments. I have expertise in building solutions for parallel processing (a basic requirement for bioinformatics analyses at industrial scale), concurrent processing (a requirement for analyses in distributed environments) and automation of large processes.

I also work on building native internet tools and public domain tools (see section Public Tools), such as Mycelium API Gateway and Mycelium WebApp, a pair of solutions focused on distributed access control of complex API environments. I can also mention data validation tools such as Blutils CLI and Blutils UI, a pair focused on biological sequence analysis and, finally, Classeq, a biological sequence classifier for post-phylogeny. All these tools, among others, are made available free of charge and open source and can be accessed and especially audited through the official GitHub of the Lepista Bioinformatics project.

Links:

[1] A. Góes-Neto; [2] E. R. Deschsler-Santos; [3] J. P. M. Araújo; [4] J. C. Dianese; [5] M. A. Neves; [6] R. W. Barreto; [7] C. W. Barnes; [8] M. C. Aime; [9] H. C. Evans; [10] D. Begerow.

TITRATION

PHD IN MICROBIOLOGY

Institution	University of Brasília (UnB)
Degree	PhD in Microbiology
Year	2017

In my thesis entitled "Fungi in Cerrado plants: an integrative approach", I developed two bioinformatics tools called "GeneConnector" and "Classeq". The first designed to connect genetic information using metadata intrinsic to GenBank information. The second, "Classeq", is a biological sequence classifier that uses pre-built phylogenies to position DNA sequences, with high performance, relatively low computational cost and low error rate, using an environment friendly to machine connections via API.

MASTER IN FUNGI, ALGAE AND PLANT BIOLOGY

Institution	Federal University of Santa Catarina (UFSC)
Degree	Master in Fungi, Algae and Plant Biology
Year	2015

In the development of my master's dissertation, I conducted a biogeographic study on the fungus Phellinotus piptadeniae, a fungus widely distributed in the neotropical region, related to legume species. On that occasion we used environmental modeling tools, taking advantage of the fungus/host relationship, to identify the probable geographic distribution of the fungus. In addition, we identified a possible species complex within Phellinotus, presenting a diffuse evolutionary history, shaped by climate fluctuations and ecological relationships.

BACHELOR IN BIOLOGICAL SCIENCES

Institution	University of the Extreme South of Santa Catarina (UNESC)
Degree	Bachelor in Biological Sciences
Year	2011

At this stage of my academic training, I acquired my strongest knowledge in pathogen-host interaction biology, using bacteria/non-human animal models in the spectrum of neurodegenerative diseases, such as meningitis. At this time I acquired my knowledge in biostatistics, n-dimensional modeling and had the opportunity to participate in research with neuroscientists Tatiane Barichello and João Quevedo, both of international renown in their respective fields of knowledge.

PUBLICATIONS IN JOURNALS

My trajectory has been marked by two major professional moments: my scientific initiation in the world of Health Sciences and my professional evolution in the world of Life Sciences and Bioinformatics. In the following sections, I list my scientific contributions organized by field of knowledge and chronological order in reverse order.

LIFE SCIENCES AND BIOINFORMATICS

In this section, I list my scientific contributions related to fields of knowledge in Life Sciences and Bioinformatics, with publications made between the years 2018 and 2024.

1. 2024: GeneConnector: Unlocking the full potential of Genbank metadata; ELIAS, S. GALVAO ; GUTERRES, D. CERVIERI ; BARRETO, R. WEINGART ; VALE, H. MARIO MARTINS DO; IEEE Latin America Transactions, vol. 22, no. 2, pp. 99-105, Feb. 2024, doi: 10.1109/TLA.2024.10412034.

2. 2022: Neotropical Studies on Hymenochaetaceae: Unveiling the Diversity and Endemicity of Phellinotus; SALVADOR-MONTOYA, C. A. ; ELIAS, S. G. ; POPOFF, O. F. ; ROBLEDO, G. L. ; URCELAY, C. ; GÓES-NETO, A. ; MARTÍNEZ, S. ; DRECHSLER-SANTOS, E. R.; J Fungi (Basel). 2022 Feb 22;8(3):216. doi: 10.3390/jof8030216. PMID: 35330218; PMCID: PMC8950266.

3. 2022: Reinstatement and phylogenetic allocation of the palm rust genus Cerradoa in the Pucciniaceae, and establishment of Pseudocerradoa, gen. nov; EBINGHAUS, MALTE ; DOS SANTOS, MARIA D. M. ; SOUZA, ERICA S. C. ; BARNES, CHARLES W. ; NDACNOU, MIRAINE K. ; VÉLEZ-ZAMBRANO, SÉRGIO M. ; GALVÃO-ELIAS, SAMUEL ; BEGEROW, DOMINIK ; BARRETO, R. W. ; Dianese, José C.; MYCOLOGIA, v. 115, p. 1-19, 2022

4. 2021: Phytophthora theobromicola sp. nov.: A New Species Causing Black Pod Disease on Cacao in Brazil; DECLOQUEMENT, J. ; RAMOS-SOBRINHO, R. ; ELIAS, S. G. ; BRITTO, D. S. ; PUIG, A. S. ; REIS, A. ; DA SILVA, R. A. F. ; HONORATO-JÚNIOR, J. ; LUZ, E. D. M. N. ; PINHO, D. B. ; MARELLI, J.-P.; Frontiers in Microbiology, Volume 12, 2021. https://www.frontiersin.org/journals/microbiology/articles/10.3389/fmicb.2021.537399. DOI: 10.3389/fmicb.2021.537399. ISSN: 1664-302X.

5. 2020: Studies on the biogeography of Phellinotus piptadeniae (Hymenochaetales, Basidiomycota): Expanding the knowledge on its distribution and clarifying hosts relationships; ELIAS, S. G. ; SALVADOR-MONTOYA, C. A. ; COSTA-REZENDE, D. H. ; GUTERRES, D. C. ; FERNANDES, M. ; OLKOSKI, D. ; KLABUNDE, G. H. F. ; DRECHSLER-SANTOS, E. R.; Fungal Ecology, Volume 45, 2020, 100912, ISSN 1754-5048.

6. 2020: Moniliophthora perniciosa, the mushroom causing witches' broom disease of cacao: Insights into its taxonomy, ecology and host range in Brazil; LISBOA, D. O. ; EVANS, H. C. ; ARAÚJO, J. P. M. ; ELIAS, S. G. ; BARRETO, R. W.; Fungal Biology, Volume 124, Issue 12, 2020,

Pages 983-1003, ISSN 1878-6146, https://doi.org/10.1016/j.funbio.2020.09.001. (https://www.sciencedirect.com/science/article/pii/S1878614620301276)

7. 2020: A new section, Lactifluus section Neotropicus (Russulaceae), and two new Lactifluus species from the Atlantic Forest, Brazil; DUQUE BARBOSA, J. A. ; DELGAT, L. ; GALVÃO ELIAS, S. ; VERBEKEN, A. ; NEVES, M. A. ; CARVALHO, A. ALVES DE; Systematics and Biodiversity, 18(4), 347–361. https://doi.org/10.1080/14772000.2020.1769221

8. 2019: Phaeochorellaceae, Diaporthales: a new fungal family and a re-appraisal of Phaeochorella species.; GUTERRES, D. C. ; ELIAS, S. G. ; SANTOS, M. D. M. ; SOUZA, B. C. P. ; ALMEIDA, C. P. ; PINHO, D. B. ; MILLER, R. N. G. ; DIANESE, J. C.; MYCOLOGIA, v. 111, p. 1-16, 2019

9. 2019: Phylogenetic Relationships of Phaeochorella Parinarii and Recognition of a New Family, Phaeochorellaceae (Diaporthales); GUTERRES, D. C. ; GALVÃO-ELIAS, S. ; SANTOS, M. D. M. ; SOUZA, B. C. P. ; ALMEIDA, C. P. ; PINHO, D. B. ; MILLER, R. N. G. ; DIANESE, J. C.; Mycologia 111, no. 4 (2019): 660–75. https://www.jstor.org/stable/26756647.

10. 2018: Taxonomy, phylogeny, and divergence time estimation for Apiosphaeria guaranitica, a neotropical parasite on bignoniaceous hosts.; GUTERRES, D. C. ; GALVAO-ELIAS, S. ; SOUZA, B. C. P. ; PINHO, D. B. ; SANTOS, M. D. M. ; MILLER, R. N. G. ; DIANESE, J. C.; MYCOLOGIA, v. 110, p. 526-545, 2018

11. 2018: Crossopsorella, a new tropical genus of rust fungi; SOUZA, E. S. C. ; AIME, M. C. ; ELIAS, S. G. ; PINHO, D. B. ; MILLER, R. N. G. ; DIANESE, J. C.; Phytotaxa, v. 375, p. 189-202, 2018

HEALTH SCIENCES

In this section, I list my scientific contributions related to fields of knowledge in Health Sciences, with publications made between the years 2013 and 2014.

1. 2014: Inhibition of matrix metalloproteinases-2 and -9 prevents cognitive impairment induced by pneumococcal meningitis in Wistar rats; BARICHELLO, T. ; GENEROSO, J. S. ; MICHELON, C. M. ; SIMÕES, L. R. ; ELIAS, S. G. ; VUOLO, F. ; COMIM, C. M. ; DAL-PIZZOL, F. ; QUEVEDO, J.; Exp Biol Med (Maywood). 2014 Feb;239(2):225-31. doi: 10.1177/1535370213508354. Epub 2014 Jan 13. PMID: 24419461.

2. 2014: Klebsiella pneumoniae meningitis induces memory impairment and increases pro-inflammatory host response in the central nervous system of Wistar rats; BARICHELLO, T. ; SIMÕES, L. R. ; VALVASSORI, S. S. ; GENEROSO, J. S. ; AVELINE, P. E. D. V. ; DOMINGUINI, D. ; ELIAS, S. G. ; VILELA, M. C. ; QUEVEDO, J. ; TEIXEIRA, A. L.; J Med Microbiol. 2014 Jan;63(Pt 1):111-117. doi: 10.1099/jmm.0.063289-0. Epub 2013 Oct 8. PMID: 24105840.

3. 2013: Inhibition of indoleamine 2,3-dioxygenase prevented cognitive impairment in adult Wistar rats subjected to pneumococcal meningitis; BARICHELLO, T. ; GENEROSO, J. S. ; SIMÕES, L. R. ; ELIAS, S. G. ; TASHIRO, M. H. ; DOMINGUINI, D. ; COMIM, C. M. ; VILELA, M. C. ; TEIXEIRA, A. L. ; QUEVEDO, J.; Translational Research, Volume 162, Issue 6, 2013, Pages 390-397, ISSN 1931-5244, https://doi.org/10.1016/j.trsl.2013.08.001. (https://www.sciencedirect.com/science/article/pii/S193152441300251X)

4. 2013: Pathophysiology of bacterial infection of the central nervous system and its putative role in the pathogenesis of behavioral changes; BARICHELLO, T. ; GENEROSO, J. S. ; MILIOLI, G. ; ELIAS, S. G. ; TEIXEIRA, A. L.; Braz J Psychiatry. 2013 Mar;35(1):81-7. doi: 10.1016/j.rbp.2012.11.003. PMID: 23567606.

5. 2013: Pathophysiology of neonatal acute bacterial meningitis; BARICHELLO, T. ; FAGUNDES, G. D. ; GENEROSO, J. S. ; ELIAS, S. G. ; SIMÕES, L. R. ; TEIXEIRA, A. L.; J Med Microbiol. 2013 Dec;62(Pt 12):1781-1789. doi: 10.1099/jmm.0.059840-0. Epub 2013 Aug 14. PMID: 23946474.

6. 2013: Role of oxidative stress in the pathophysiology of pneumococcal meningitis; BARICHELLO, T. ; GENEROSO, J. S. ; SIMÕES, L. R. ; ELIAS, S. G. ; QUEVEDO, J.; Oxid Med Cell Longev. 2013;2013:371465. doi: 10.1155/2013/371465. Epub 2013 May 9. PMID: 23766853; PMCID: PMC3665263.

PUBLIC DOMAIN SOFTWARE

Nos últimso três anos venho trabalhando de maneira independente no desenvolvimento de ferramentas de bioinformática e nativas da web, na forma de soluções open source e gratuitas. O objetivo é contribuir para a comunidade científica e a sociedade em geral, com ferramentas que entreguem real impacto a cientistas e não cientistas. Entre as ferramentas desenvolvidas estão:

Mycelium API Gateway

An open and free API Gateway, designed to operate in modern, multi-tenant, and API-oriented environments. The project prioritizes architectural clarity, security, and extensibility, maintaining an explicit separation between technical concerns and organizational aspects of the project.

GitHub Documentation

Mycelium WebApp

The official web interface of Mycelium API Gateway, designed to allow rapid implementation of access policies to the API Gateway endpoints, without the need for additional implementation.

GitHub

Blutils CLI

A command line interface designed to be an Blast Wrapper, used to improve their execution performance and implements an exclusive consensus algorithm to improve the interpretation of the results.

GitHub Documentation Crates

Blutils UI

A web interface designed to visualize the results of the Blutils CLI. It is designed to be used in the browser, without the need for additional implementation.

GitHub Online Interface

Classeq

A biological sequence classifier designed to perform the biological sequences placement into a predefined phylogeny. It is designed as a web friendly application, with a focus on usability.

GitHub Documentation

PRIVATE SOFTWARE REGISTRATIONS

Currently, my main focus is on developing bioinformatics tools for the Agrobiotechnology area, with the development of tools for genomic and metagenomic data analysis, as well as tools for biological sequence data analysis. Additionally, I have been working voraciously on building interfaces and web-friendly high-availability tools used in delivering software solutions for Brazilian agribusiness.

Below are listed the software registered between the years 2022 and 2023. These range from software/pipelines for processing metagenomic data to high-availability web tools used in delivering software solutions for Brazilian agribusiness. In addition to the listed software, I am working on registering eleven newly developed software to be registered for the year 2026, which will soon be made available for publicly at the Brazilian Patent and Trademark Office (INPI).

Year	Title	Kind (Language)	Application
2022	AgroBase-Python	API (Python)	BR 51 2022 003471 9
-	-	API (Python)	BR 51 2022 003472 7
-	QuorumSensing	API/ETL (Python)	BR 51 2022 003473 5
2023	AgroBase-Python	LIBRARY (Python)	BR 51 2023 003286 7
-	AgroBase-Rust	LIBRARY (Rust)	BR 51 2023 003288 3
-	AgroIndexAPI	API (Python)	BR 51 2023 003300 6
-	Agroportal	WEB (Typescript/HTML/CSS)	BR 51 2023 003302 2
-	AgroReporterUI	WEB (Typescript/HTML/CSS)	BR 51 2023 003305 7
-	BioArchival	API (Rust)	BR 51 2023 003309 0
-	BioReferee	API (Python)	BR 51 2023 003311 1
-	BioTax	API (Rust)	BR 51 2023 003321 9
-	MetaDaVinci	Pipeline (Python)	BR 51 2023 003324 3
-	PathFinder	Pipeline (Python)	BR 51 2023 003327 8
-	QuorumSensing	API/ETL (Python)	BR 51 2023 003328 6